Aadit Kshirsagar

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EDUCATION

University of Southern California — Aug 2025 - May 2027
Master of Science in Computer Science – Artificial Intelligence

Savitribai Phule Pune University — Aug 2020 - Jun 2024
Bachelor of Engineering in Computer Science — *GPA:8.89/10*

TECHNICAL SKILLS

- **Programming**: Python, Java, SQL, Git
- **Databases**: PostgreSQL, MongoDB, Chroma, Pinecone, Milvus, Weaviate
- **Web Technologies**: HTML, CSS, JavaScript, Streamlit
- **Cloud and DevOps**: Azure DevOps, Google Cloud, Amazon Web Services, Fast API, Docker, Kubernetes, Azure AI, Open AI, Amazon Bedrock
- **AI/ML**: Generative AI, NLP, LLMOps, MLOps, RAG, AutoGen, MCP, LangChain, LlamaIndex, LangGraph, Prompt Engineering, Computer Vison, Transformers, LoRa, Pandas, PySpark, PyTorch, TensorFlow, Keras

WORK EXPERIENCE

Calsoft Pvt Ltd | Data Science Intern | Pune, India — Apr 2024 - Jun 2025
- Executed end-to-end project lifecycle within core GenAI team, from architecture to optimization, delivering scalable, AI-driven solutions.
- Built a multi-pipeline, agent-based architecture, increasing test case relevance accuracy to 95% and enabling retrieval of legacy test cases.
- Developed and deployed on-premises AI solution with open-source LLMs, achieving 80% reduction in manual test case generation and enhancing QA productivity.

Intellect Design Arena Ltd | AI/ML Intern | Pune, India — Sep 2023 - Mar 2024
- Facilitated development of IGTB Copilot, an internal project enabling Relationship Managers to leverage actionable insights from bank data.
- Implemented LLM integration and prompt engineering for a successful proof of concept Copilot, automating payment onboarding workflows by over 70%.

Intellect Design Arena Ltd | Software Developer Intern | Pune, India — Jan 2023 - Mar 2023
- Engineered Natural Language Processing techniques for efficient filtering of bank customer data.
- Developed a project utilizing Streamlit and AI tools, resulting in a 45% increase in efficiency of KYC checks for bank customers.

PROJECTS

Text Gen from Scratch | Codebase
- Engineered a 6-layer transformer with custom positional encoding and learning rate scheduling to improve training convergence.
- Built an end-to-end ML pipeline for text generation with a 50k-word vocabulary and 64-token sequences, enabling scalable training across 50 epochs.

Medi Match Copilot | Codebase
- Integrated Llama 3.1 for natural language processing and Llama 3.2-Vision for medical image analysis, supporting robust multimodal reasoning in clinical workflows and achieving up to 95% confidence in diagnostic assessments.
- Created a modular Python-based system for scalable retrieval of medical data, generation of diagnostic hypotheses, and visual analysis, tailored for long-term maintainability.
- Delivered a comprehensive clinical assistant automating patient symptom interpretation, medical image evaluation, and evidence-based knowledge synthesis, assisting in data-informed diagnostic decisions.

Eco Policy Copilot | Codebase
- Integrated advanced data modeling and workflow automation to support evidence-based decision-making for climate and economic policy initiatives, assisting environmental researchers and analysts.
- Constructed an AI-driven platform using LangChain and LangGraph, processing 6 data sources and over 20 policy types, analyzing climate, economic, and energy data, and delivering actionable policy recommendations for sustainable development.
- Designed an Agentic workflow, enabling autonomous agents to process data, simulate policies, and generate executive summaries for streamlined environmental analysis.

PUBLICATIONS

- "**Enhancing RAG performance through Chunking and Text Splitting Techniques"** published in the International Journal of Scientific Research in Computer Science, Engineering, and Information Technology. | Link